

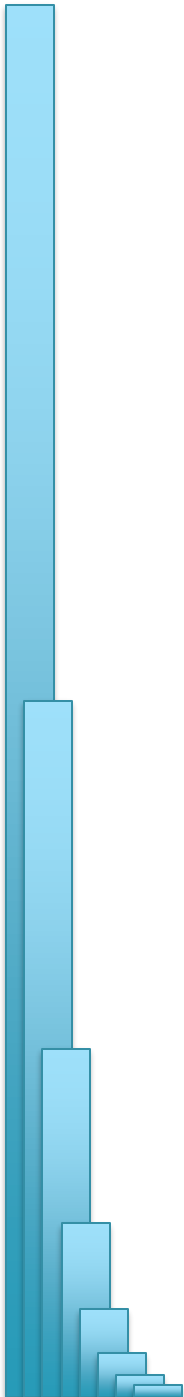
Human Genome & Read Mapping

Michael Schatz

Sept 10, 2024

Lecture 5: Applied Comparative Genomics

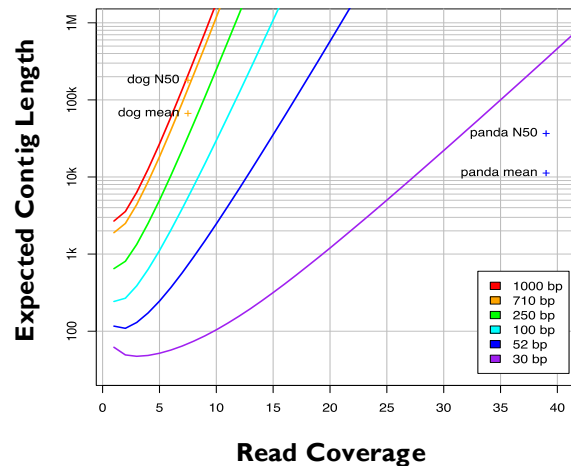




Recap & Assignment 2

Ingredients for a good assembly

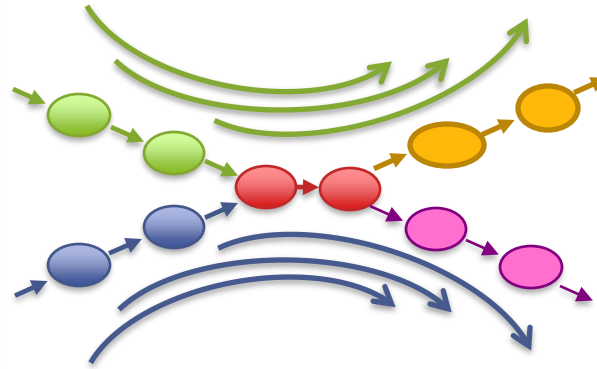
Coverage



High coverage is required

- Oversample the genome to ensure every base is sequenced with long overlaps between reads
- Biased coverage will also fragment assembly

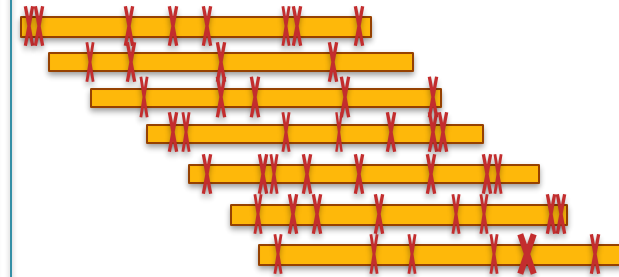
Read Length



Reads & mates must be longer than the repeats

- Short reads will have **false overlaps** forming hairball assembly graphs
- With long enough reads, assemble entire chromosomes into contigs

Quality



Errors obscure overlaps

- Reads are assembled by finding kmers shared in pair of reads
- High error rate requires very short seeds, increasing complexity and forming assembly hairballs

Current challenges in *de novo* plant genome sequencing and assembly

Schatz MC, Witkowski, McCombie, WVR (2012) *Genome Biology*. 12:243

1. **Coverage**: low coverage is mathematically hopeless
2. **Repeat composition**: high repeat content is challenging
3. **Read length**: longer reads help resolve repeats
4. **Error rate**: errors reduce coverage, obscure true overlaps

- Assembly is a hierarchical, starting from individual reads, build high confidence contigs/unitigs, incorporate the mates to build scaffolds
 - Extensive error correction is the key to getting the best assembly possible from a given data set
- Recommend spades for short read assembly
 - Integrates error correction and scaffolding

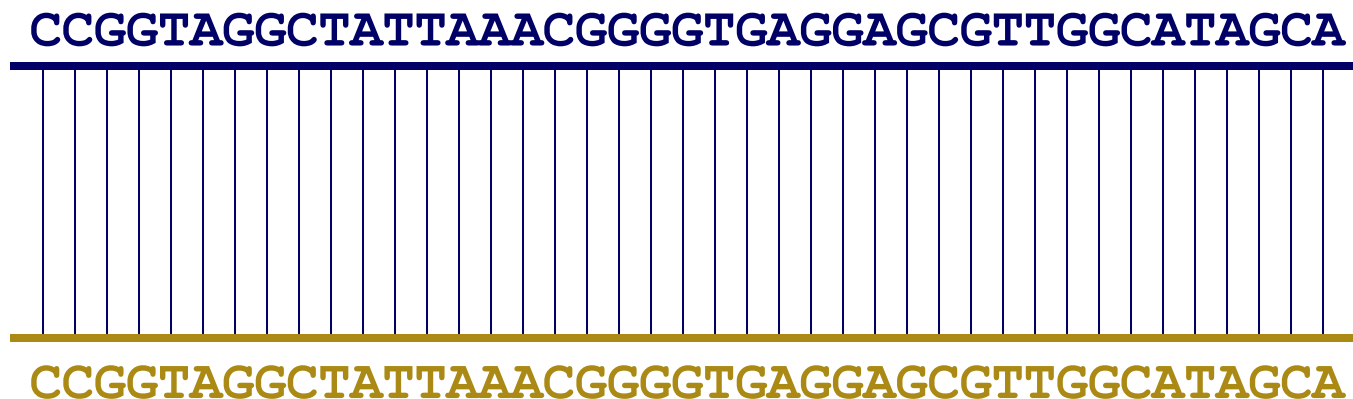


Whole Genome Alignment with MUMmer

Slides Courtesy of Adam M. Phillippy
NHGRI

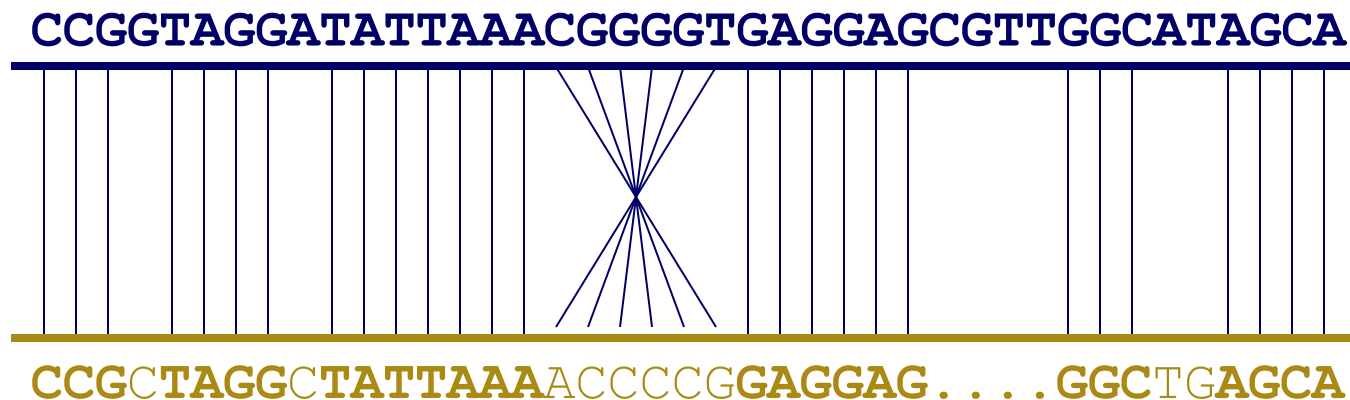
Goal of WGA

- For two genomes, A and B , find a mapping from each position in A to its corresponding position in B



Not so fast...

- Genome *A* may have insertions, deletions, translocations, inversions, duplications or SNPs with respect to *B* (sometimes all of the above)



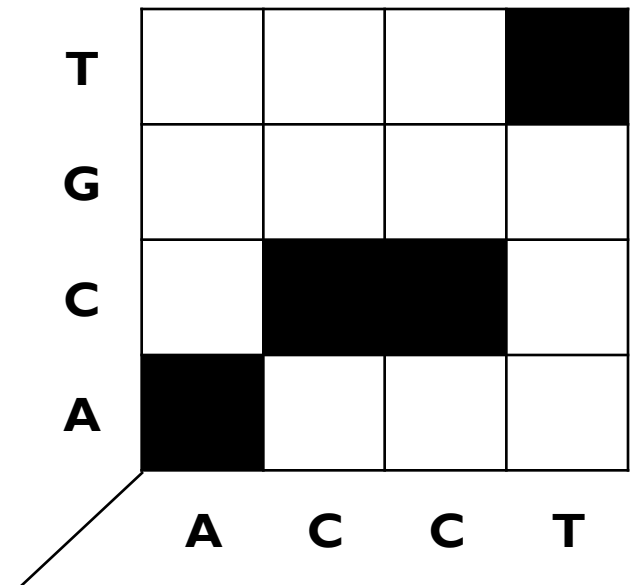
WGA visualization

- How can we visualize *whole* genome alignments?

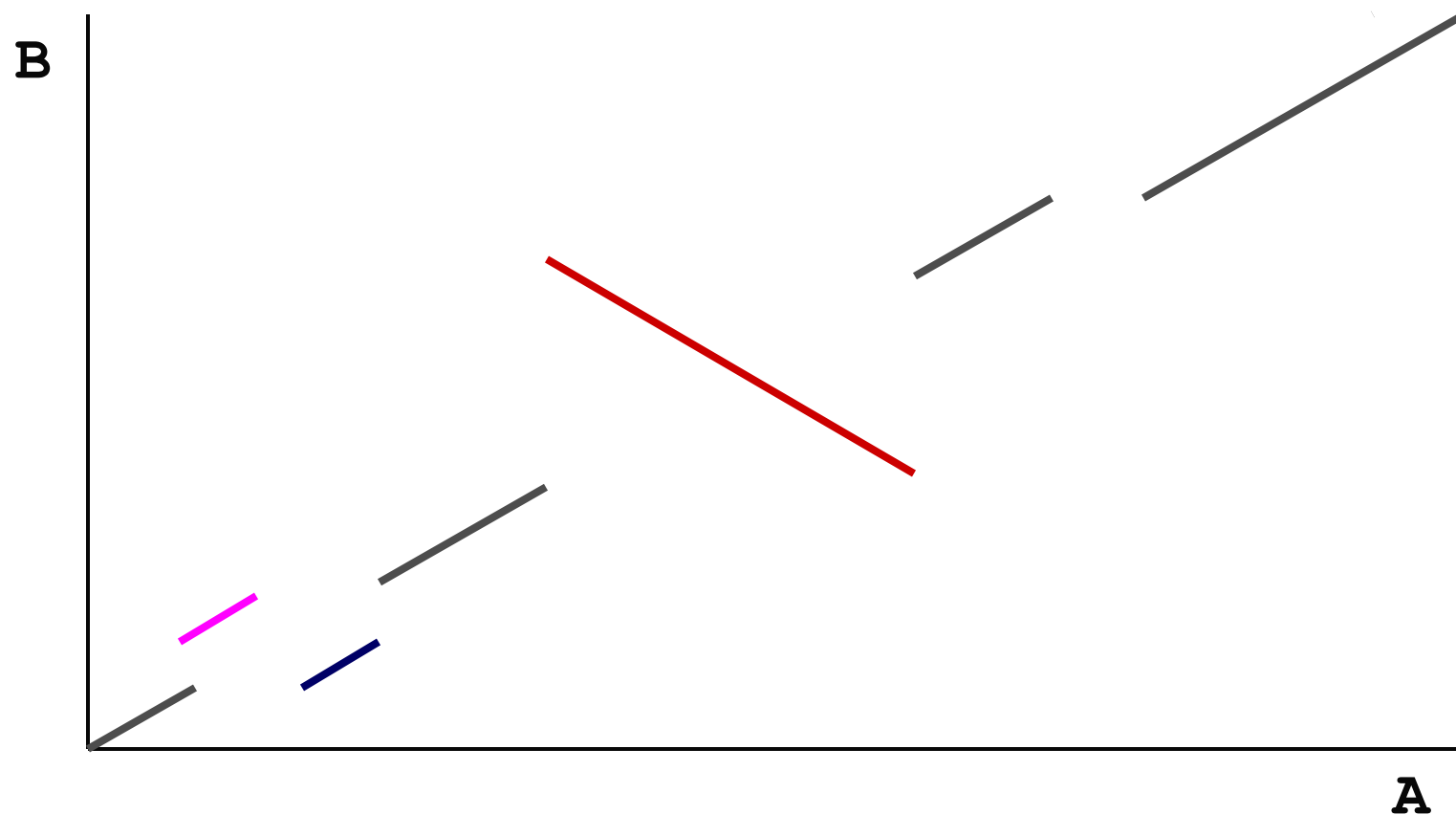
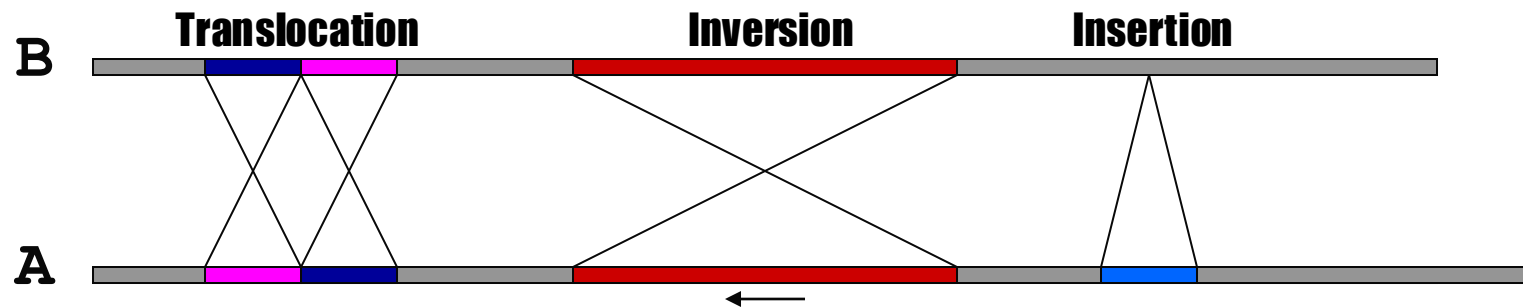
- With an alignment dot plot

- $N \times M$ matrix

- Let i = position in genome A
 - Let j = position in genome B
 - Fill cell (i,j) if A_i shows similarity to B_j



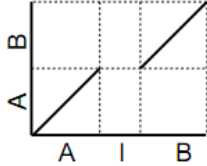
- A perfect alignment between A and B would completely fill the positive diagonal



SV Types

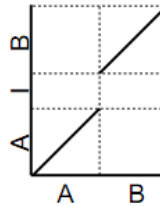
Insertion into Reference

R: AIB
Q: AB



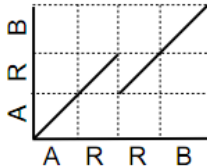
Insertion into Query

R: AB
Q: AIB



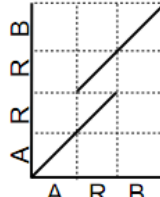
Collapse Query

R: ARRB
Q: ARB



Collapse Reference

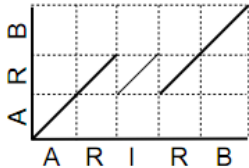
R: ARB
Q: ARRB



Collapse Query
w/ Insertion

R: ARIRB
Q: ARB

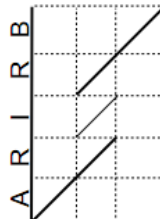
Exact tandem
alignment if I=R



Collapse Reference
w/ Insertion

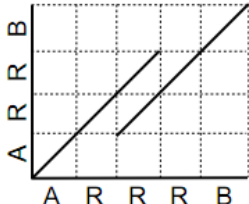
R: ARB
Q: ARIRB

Exact tandem
alignment if I=R



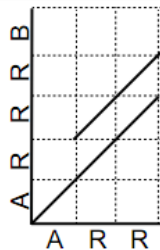
Collapse Query

R: ARRRB
Q: ARRB



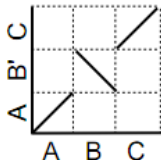
Collapse Reference

R: ARRB
Q: ARRRB



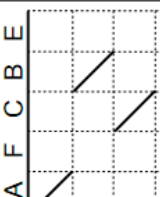
Inversion

R: ABC
Q: AB'C

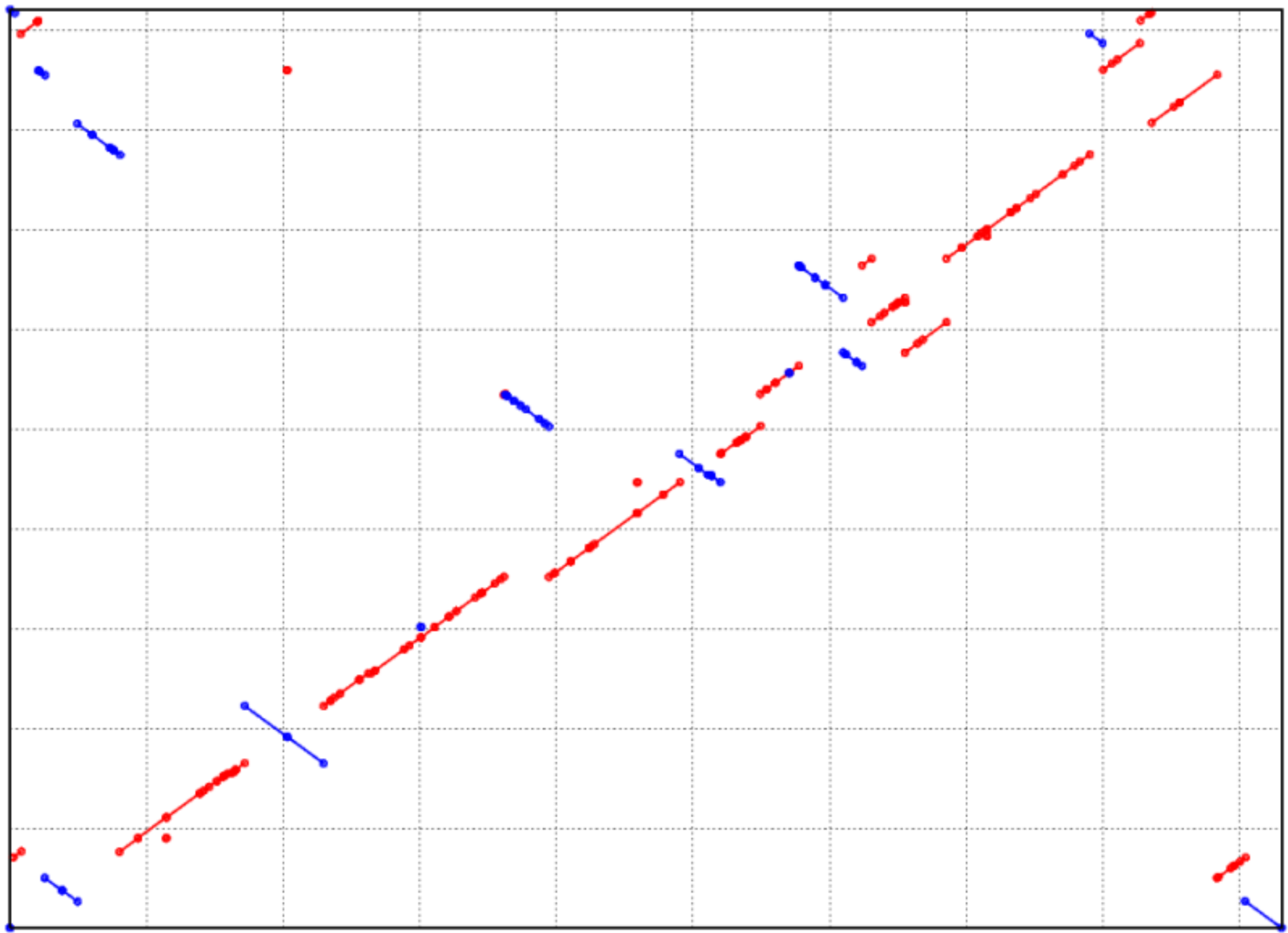


Rearrangement
w/ Disagreement

R: ABCDE
Q: AFCBE



- Different structural variation types / misassemblies will be apparent by their pattern of breakpoints
- Most breakpoints will be at or near repeats
- Things quickly get complicated in real genomes

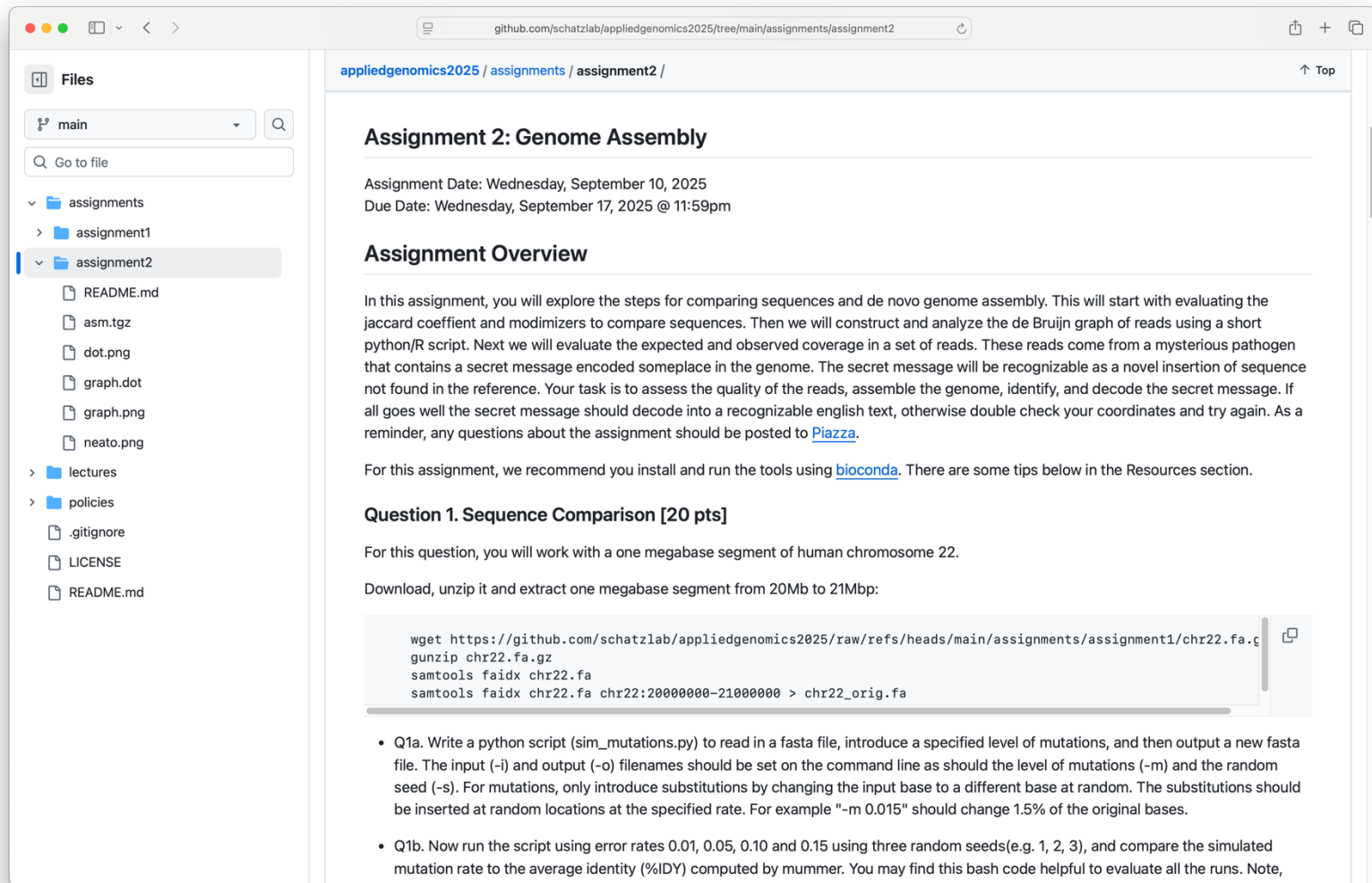


Alignment of 2 strains of *Y. pestis*

<http://mummer.sourceforge.net/manual/>

Assignment 2: Genome Assembly

Due Wed Sept 17 by 11:59pm



The screenshot shows a web browser displaying the GitHub repository page for 'appliedgenomics2025/assignments/assignment2'. The left sidebar shows the file structure with folders 'assignments', 'assignment1', and 'assignment2'. The 'assignment2' folder is selected, showing files like 'README.md', 'asm.tgz', 'dot.png', 'graph.dot', 'graph.png', 'neato.png', 'lectures', 'policies', '.gitignore', 'LICENSE', and 'README.md'. The main content area displays the 'Assignment 2: Genome Assembly' page. It includes the assignment date (Wednesday, September 10, 2025), the due date (Wednesday, September 17, 2025 @ 11:59pm), and an 'Assignment Overview' section. The overview describes the task: comparing sequences and de novo genome assembly. It mentions evaluating the jaccard coefficient and modimizers, constructing a de Bruijn graph, and analyzing coverage. A secret message is hidden in the genome. The task is to assess the quality of the reads, assemble the genome, identify, and decode the secret message. A reminder to post questions on Piazza is included. Below the overview, there is a section for 'Question 1. Sequence Comparison [20 pts]' which involves working with a one megabase segment of human chromosome 22. It provides instructions to download, unzip, and extract a segment from 20Mb to 21Mb. A code block shows the commands to download the file, unzip it, and extract the segment. The code is:

```
wget https://github.com/schatzlab/appliedgenomics2025/raw/refs/heads/main/assignments/assignment1/chr22.fa.gz
gunzip chr22.fa.gz
samtools faidx chr22.fa
samtools faidx chr22.fa chr22:20000000-21000000 > chr22_orig.fa
```

 Below the code block, there are two questions: Q1a. Write a python script (sim_mutations.py) to read in a fasta file, introduce a specified level of mutations, and then output a new fasta file. The input (-i) and output (-o) filenames should be set on the command line as should the level of mutations (-m) and the random seed (-s). For mutations, only introduce substitutions by changing the input base to a different base at random. The substitutions should be inserted at random locations at the specified rate. For example "-m 0.015" should change 1.5% of the original bases. Q1b. Now run the script using error rates 0.01, 0.05, 0.10 and 0.15 using three random seeds (e.g. 1, 2, 3), and compare the simulated mutation rate to the average identity (%IDY) computed by mummer. You may find this bash code helpful to evaluate all the runs. Note, the code is:

```
for seed in 1 2 3; do
  for error in 0.01 0.05 0.10 0.15; do
    python sim_mutations.py -i chr22_orig.fa -o chr22_${error}_${seed}.fa -m $error -s $seed
  done
done
```

<https://github.com/schatzlab/appliedgenomics2025/tree/main/assignments/assignment2>

Check Piazza for questions!

Halomonas sp. GFAJ-1



Library 1: Fragment

Avg Read length: 100bp

Insert length: 180bp

Library 2: Short jump

Avg Read length: 50bp

Insert length: 2000bp

A Bacterium That Can Grow by Using Arsenic Instead of Phosphorus

Wolfe-Simon et al (2010) *Science*. 332(6034)1163-1166.

Digital Information Storage

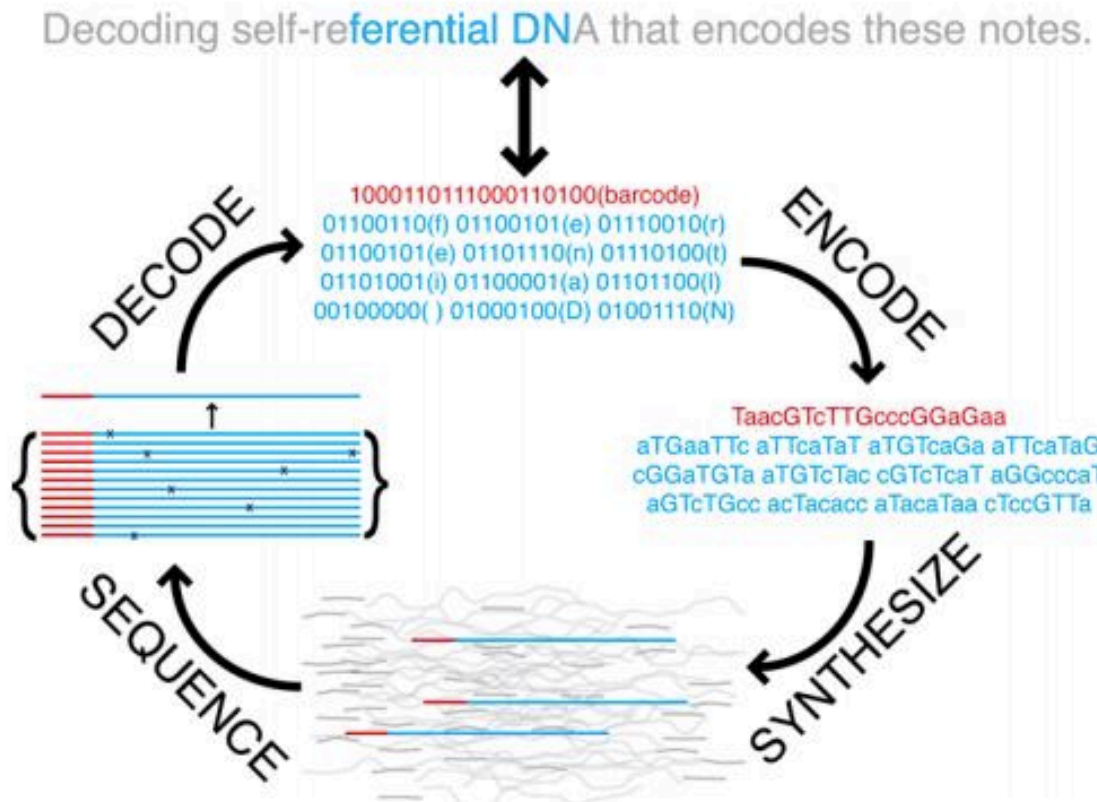


Fig. S1. Schematic of DNA information storage.

Encoding/decoding algorithm implemented in dna-encode.pl from David Dooling.

Next-generation Digital Information Storage in DNA

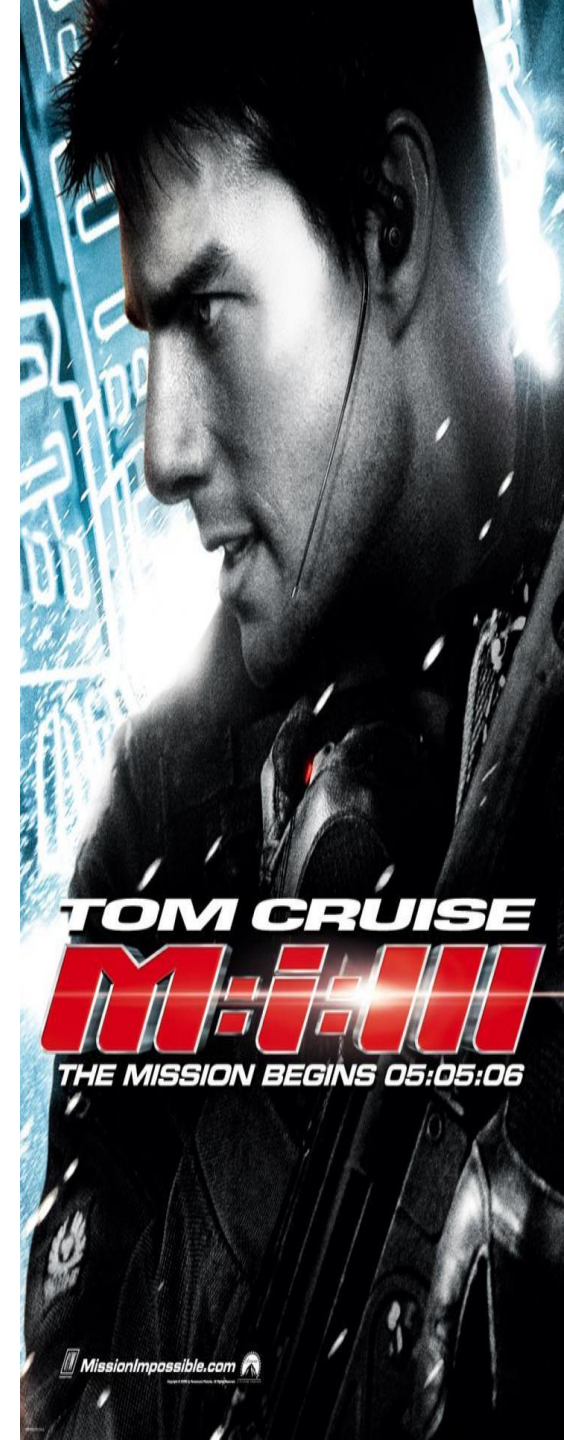
Church et al (2010) *Science*. 337(6102)1628

Assignment 2: Genome Assembly

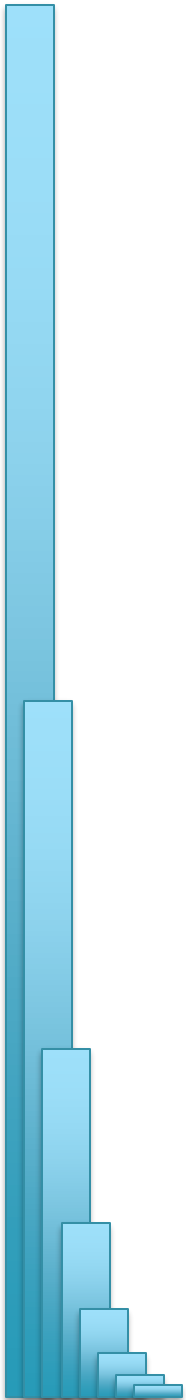
Due Wednesday Sept 17 @ 11:59pm

- 1. Setup Conda/Docker/Ubuntu**
- 2. Initialize Tools**
- 3. Download Reference Genome & Reads**
- 4. Decode the secret message**
 - Check kmer distribution
 - Assemble the reads with spades
 - Align to reference with MUMmer
 - Extract foreign sequence
 - `dna-decode.py`

<https://github.com/schatzlab/appliedgenomics2025/blob/master/assignments/assignment2/README.md>



The human genome



The scale of DNA in our body is staggering.

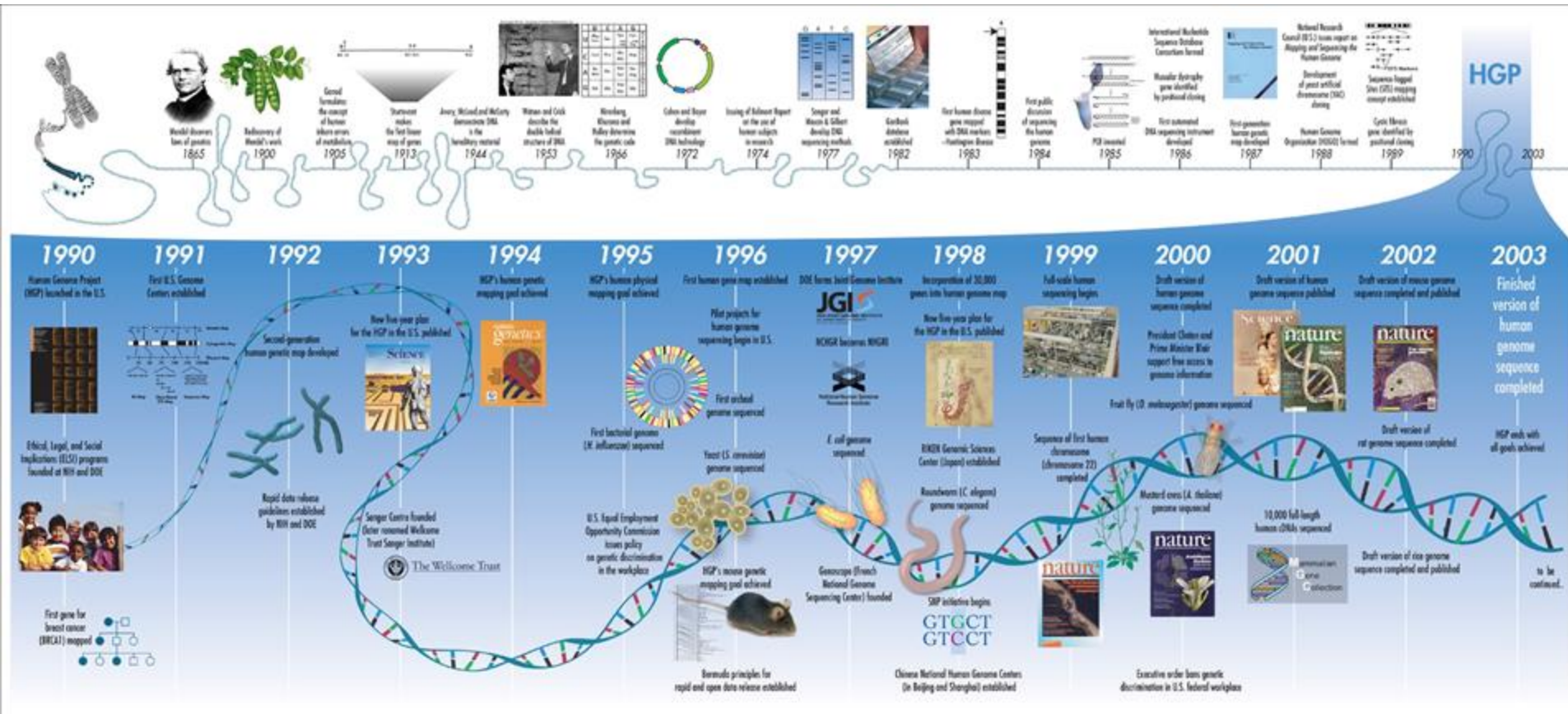
- A typical human is comprised of roughly 40 trillion human cells (excluding trillions of bacterial cells in our gut)
- If stretched out, each haploid genome would be roughly 2 meters.
- So, each cell has 4 meters of DNA.
- 40 trillion * 4 meters = 160 trillion meters.
- 160 trillion meters / 1609.34 = 99,750,623,441 miles
- 99,750,623,441 / 92,960,000 = 1,073.05 trips to the sun.

A typical cell replicates about 100 times

160 trillion meters x 100 =

1.69123746 light years

History of the Human Genome Project



The reference human genome

A photograph of Bill Clinton speaking at a podium. He is wearing a dark suit, white shirt, and blue tie. Two microphones are in front of him. In the background, two other men are visible but out of focus. The setting appears to be a formal event, possibly in a government building.

“Without a doubt, this is the most important, most wondrous map ever produced by humankind.”

*Bill Clinton
June 26, 2000*

The reference human genome

A photograph of three men standing behind a podium. In the center is Bill Clinton, President of the United States, wearing a dark suit and a blue tie. To his left is James Watson, wearing a dark suit and a red tie. To his right is Francis Collins, wearing a dark suit, a blue shirt, and a patterned tie. They are all looking towards the right. The podium has a microphone and the Presidential Seal. The background is a blurred interior with warm lighting.

“Without a doubt, this is the most important, most wondrous map ever produced by humankind.”

*Bill Clinton
June 26, 2000*



The Sequence of the Human Genome

Venter et al.

Science 291, pp 1304-1351 (2001)



Initial sequencing and analysis of the human genome

International Human Genome Sequencing Consortium

Nature 409, pp 860-921 (2001)

Who is the reference human?

The Buffalo News/Sunday, March 23, 1997

ment abuse, civil disobedience

ple. But the very nature of gov- creates a mind set that inspires increase their authority, always at ense of the people."

ly, government has forgotten that rant of the people," Parlato add- acting more like it's the master."

to and the Lapps share an abiding non-violent civil disobedience. insist on being respectful in our of resistance," Barbara Lyn Lapp but if we claim to care about our rs, we must protest government in-

violence has to be the watchword, said, calling civil disobedience the is of the violent militia movement. on-violence can serve as an anti-government oppression, he added.

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Rachel Lapp says she believes govern- ment can be good, when it controls the aggressors in society. Instead, it too often comes down on the side of the aggressors, who enforce child-protection laws, compulsory education, disclosure rules on tax forms and seat belt laws.

"We want people to see the correlation between what happened to us and what can happen to anyone when government gets out of hand," Rachel Lapp said.

The Lapps and Parlato will be joined by Samuel Radford III, a critic of public education who was arrested and pleaded guilty to reduced charges following a 1993 disturbance at the City Campus of Erie Community College.

WANTED

20 Volunteers

to participate in the

Human Genome Project

a very large international scientific research effort.

The goal is to decode the human hereditary information (*human blueprint*) that determines all individual traits inherited from parents. The outcome of the project will have tremendous impact on future progress of medical science and lead to improved diagnosis and treatment of hereditary diseases.

Volunteers will receive information about the project from the Clinical Genetics Service at Roswell Park, and sign a consent form before participating.

No personal information will be maintained or transferred.

Volunteers will provide a one-time donation of a small blood specimen. A small monetary reimbursement will be provided to the participants for their time and effort.

Individuals must be at least 18 years of age.

Persons who have undergone chemotherapy are not eligible.



For more information please contact the
Clinical Genetics Service
845-5720 (9:00 am - 3:00 pm)
March 24 - 26, 1997

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CANCER INSTITUTE

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7 May 2010 | \$10

Science



The Neandertal Genome

AAAS

Pieter de Jong, RPCI

Who is the reference human?

The Buffalo News/Sunday, March 23, 1997

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845-5720 (9:00 am - 3:00 pm)
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Appendix: Identifying the ancestry of segments of the human genome reference sequence

To compare Neandertal to present-day human haplotypes for the purpose of population genetic analysis, we needed to have long haploid sequences from present-day humans that were of known ancestry. To identify such segments, we took advantage of the fact that the human reference sequence is haploid over scales of tens of kilobases, because it is comprised of a tiling-path of Bacterial Artificial Chromosomes (BACs) or other clone types that are of typical size 50-150 kb (S92). We do not know of any other substantial source of high quality human haploid sequences of the requisite size.

Determining the ancestries of the libraries in the human genome reference sequence using HAPMIX

It is crucial to know the 'ancestry' of a clone to use it in a meaningful population genetic analysis. In what follows, we define 'ancestry' as the geographic region in which a clone's ancestor lived 1,000 years ago, inferred based on its genetic proximity to other individuals from that region today. This definition allows us to classify clones from Chinese Americans as "East Asian," from European Americans as "European," and from African Americans as either "West African" or "European".

To identify the ancestries of the libraries comprising most of the human genome reference sequence, we used a list of 26,558 clones tiling the great majority of the genome, most of which we were able to assign to a library of origin. Restricting to the autosomes, we identified 21,156 clones that seemed to fall into 9 libraries based on the naming scheme: CTA (n=199), CTB (n=356), CTC (n=452), CTD (n=1,426), RPCI-1 (n=740), RPCI-3 (n=456), RPCI-4 (n=716), RPCI-5 (n=802) and RPCI-11 (n=16,009). (In a subsequent re-examination, we identified additional clones that we likely could have classified into libraries, including 953 from RPCI-11, 632 from RPCI-1, and 490 from another library RPCI-13.) The median span of the 21,156 clones we analyzed was 112 kb, and 80% are >50kb in size. About 2/3 came from a single library, **RPCI-11**.

1. **RPCI-11 is an African American:** RPCI-11, the individual who contributed most of the human genome reference sequence, is consistent with having African American ancestry, with 42% of the clones of confident West African ancestry and 42% of the clones of confident European ancestry, and the ancestry of the remaining clones less confidently inferred. The finding of likely African American ancestry for RPCI-11 was previously reported in a study of the ancestry of RPCI-11 clones spanning the Duffy blood group locus (S93), and here we confirm this finding, and also expand the inference to the whole genome.
2. **CTD is an East Asian:** The majority of clones from CTD, the second largest library in its contribution to the human genome sequence, is likely an East Asian. In a HAPMIX analysis with CEU (European) - CHB+JPT (East Asian) as the proposed ancestral populations, the majority of clones are of confident East Asian origin, and there is no secondary mode of confident European ancestry, as might be expected from a Latino or South Asian individual.
3. **The remaining 7 libraries are European:** The remaining libraries (CTA, CTB, CTC, RPCI-1, RPCI-3, RPCI-4 and RPCI-5) are inferred to be of European ancestry, since they all have consistent distributions of inferred clone ancestries, with the majority of clones of confident European ancestry in both our HAPMIX analyses and no secondary modes.

A Draft Sequence of the Neandertal Genome

Green et al (2010) Science. DOI: 10.1126/science.1188021
Supplemental Note 16 (pg 145-146)

Pieter de Jong, RPCI

Who is the reference human?

nature **methods**
Techniques for life scientists and chemists

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EDITORIAL
Nature Methods **7**, 331 (2010)
doi:10.1038/nmeth0510-331
E pluribus unum
If the human reference genome is to reflect more of the actual genomic diversity in humans, community participation is needed.
[Please visit methagora to view and post comments on this article.](#)

The human genome is ten years old. We acknowledge its reference assembly as an invaluable resource essential for many purposes such as the assembly of short reads from high-throughput sequencing platforms into chromosome context during resequencing projects. At the same time, we think necessary improvement of the reference genome depends on the willingness of the research community to provide data for the genome's less accessible regions.

First published in 2001, the human reference genome has, since 2007, been in the hands of the Genome Reference Consortium (GRC) a small group of fewer than 20 scientists from the European Bioinformatics Institute, the US National Center for Biotechnology Information, The Sanger Institute and The Genome Center at Washington University in St. Louis, who have committed to the improvement and completion of this reference, with very little financial support.

The reference genome is now in its 19th rendition, and probably the best measure of its improvement over the last ten years is the number of fragments it consists of. The very first version had ~150,000 gaps; the most recent build, GRCh37, has only around 250 gaps.

The only other publicly accessible *de novo* assembly of a human genome that contains chromosome sequences is HuRef. Obtained by traditional capillary sequencing, HuRef is the diploid genome of Craig Venter. It comes in 4,500 pieces and, like any individual genome, it contains many rare alleles.

GRCh37, in contrast, is a mosaic haploid genome derived from about 13 people. It still contains rare alleles, but the GRC recently decided to convert these to common haplotypes. Deciding which alleles are common and which are rare is proving challenging, and the GRC members are collaborating with members of the 1000 Genomes project to collect enough data to make these decisions.

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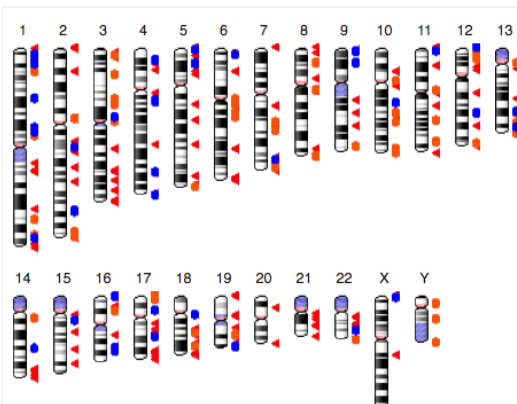
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Human Genome Overview

Information about the continuing improvement of the human genome



- ◀ Region containing alternate loci
- Region containing fix patches
- Region containing novel patches

Ideogram of the latest human assembly, GRCh38.p11

[GRCh38.p11](#)
[GRCh37.p13](#)
[GRCh37](#)

GRCh38.p11

Release date: June 14, 2017

Release type: minor

Release notes: GRCh38.p11 is the eleventh patch release for the GRCh38 reference assembly. No chromosome coordinate of patch scaffolds is now: 64 FIX and 59 NOVEL.

Assembly accessions: GenBank: [GCA_000001405.26](#) , RefSeq: [GCF_000001405.37](#)

Pseudoautosomal regions

Name	Chr	Start	Stop
PAR#1	X	10,001	2,781,479
PAR#2	X	155,701,383	156,030,895
PAR#1	Y	10,001	2,781,479
PAR#2	Y	56,887,903	57,217,415

The GRC is working hard to provide the best possible representation of the human genome by both generating multiple representations (alternates) and by providing a single path. Additionally, we are releasing alternate loci to allow users who are interested in a specific locus to access the data without affecting users who need chromosome coordinate space.

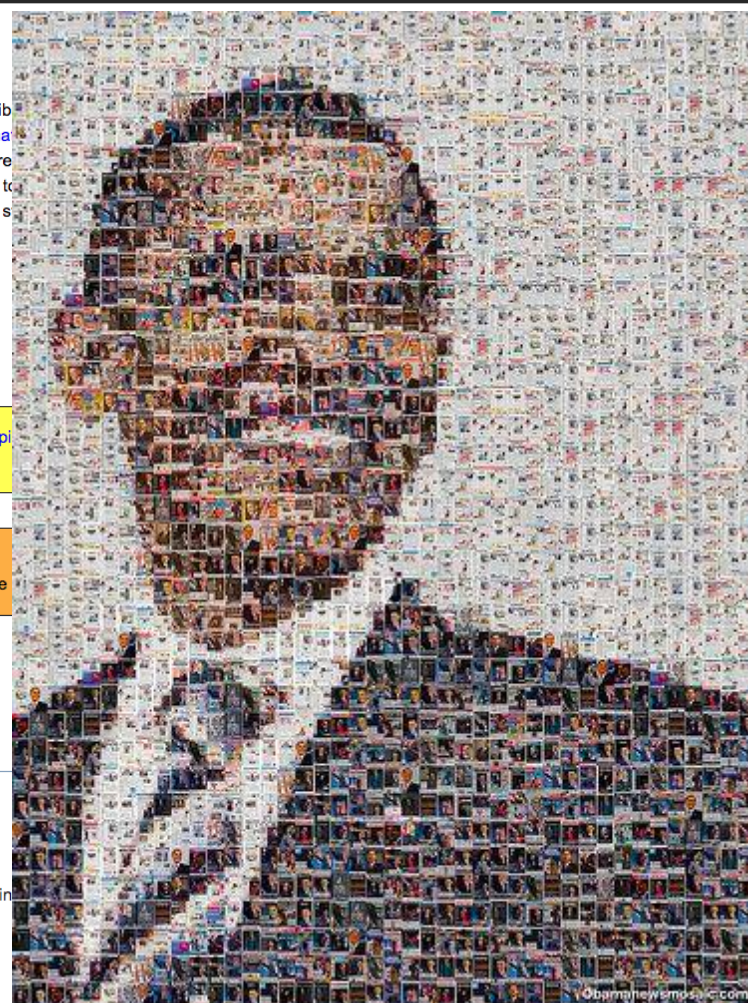
Download data:

- [GRCh38.p11 \(latest minor release\) FTP](#)
- [GRCh38 \(latest major release\) FTP](#)
- [Genomic regions under review FTP](#)
- [Current Tiling Path Files \(TPFs\)](#)

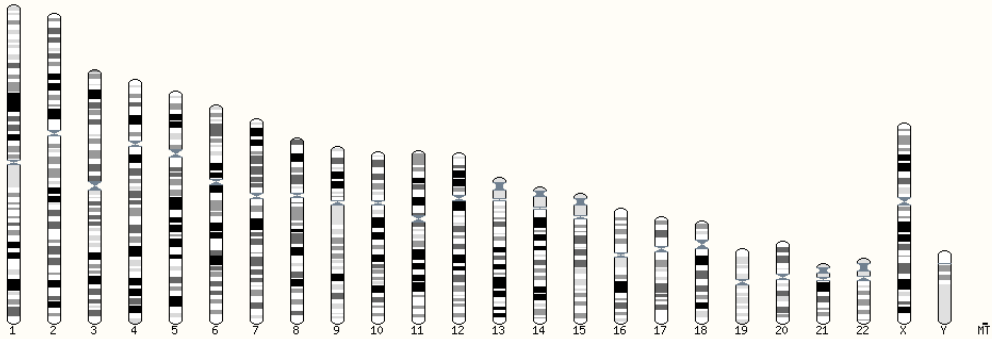
Transitioning to GRCh38? Try the [NCBI Remapping tool](#) to view the assembly alignments used by the GRC.

Next assembly update

The next assembly update (GRCh38.p12) will be released in the near future.



The human genome - basic stats



- 3.096 billion base pairs (haploid)
- 20,454 protein coding genes
- 226,950 coding transcripts
(isoforms of a gene that each encode a distinct protein product)

Assembly	GRCh38.p12 (Genome Reference Consortium Human Build 38), INSDC Assembly GCA_000001405.27 , Dec 2013
Base Pairs	3,609,003,417
Golden Path Length	3,096,649,726
Annotation provider	Ensembl
Annotation method	Full genebuild
Genebuild started	Jan 2014
Genebuild released	Jul 2014
Genebuild last updated/patched	Mar 2019
Database version	97.38
Gencode version	GENCODE 31

Gene counts (Primary assembly)

Coding genes	20,454 (incl 660 readthrough)
Non coding genes	23,940
Small non coding genes	4,871
Long non coding genes	16,848 (incl 302 readthrough)
Misc non coding genes	2,221
Pseudogenes	15,204 (incl 8 readthrough)
Gene transcripts	226,950